Business Finance Midterm Cheat Sheet: Ch.2–7

Formulas · BA II Plus keystrokes · Key concepts | Prof. Gemma Lee, Kyung Hee Univ. | Lecture-slide scope only

0. BA II Plus Setup

0.1 Reset (before every problem)

- `2nd CLR TVM` clears N, I/Y, PV, PMT, FV (CE|C, ON/OFF, 2nd QUIT do **not**)
- Cash-flow (NPV) problems: `2nd CLR WORK`
- P/Y = 1: `2nd P/Y` `1 ENTER` `↓` `1 ENTER` `2nd QUIT`
- 4 decimals: `2nd FORMAT` `4 ENTER` `2nd QUIT`
- BGN on screen → switch to END (coupons and ordinary annuities pay at period end)
- Full reset (`2nd RESET ENTER`) sets P/Y back to 12 → reset P/Y = 1

0.2 Input rules

- Enter 4 of N, I/Y, PV, PMT, FV; `CPT` + the 5th
- Cash out = **−**, cash in = **+**; same sign on PV and FV → Error 5
- I/Y = rate per period as a % number (1% → 1)
- Single sum → PMT = 0
- Negative answer = amount to pay now, not an error
- Answer off by exactly (1 + r) → BGN mode is on

Ch.2 Financial Statements, Taxes and Cash Flow

2.1 Balance Sheet

- Snapshot at a point in time; **Assets = Liabilities + Equity**
- Assets in decreasing liquidity; liabilities by when due
- NWC = current assets − current liabilities
- Liquidity = speed and ease of conversion to cash; avoids financial distress

2.2 Book vs. Market Value; Equity Accounts

- Book = accounting value (historical cost); market = price in actual trade → use market value for decisions
- Common stock = shares × par; paid-in surplus = (issue price − par) × shares
- Equity = paid-in capital + retained earnings; par = arbitrary nominal amount
- Ex: 100 sh, par $1, issued at $6 → $100 + $500 = **$600**

2.3 Income Statement

- Performance over a period; sales first, then subtract costs
- Sales − COGS − Depreciation = **EBIT**; EBIT − Interest = taxable income; − Taxes = **NI**
- NI = dividends + addition to retained earnings
- Ex: 1,509 − 750 − 89 = 670; 670 − 70 = 600; tax 21% = 126; NI = **474**
- Trap: interest is deducted **after** EBIT

2.4 GAAP vs. IFRS

- GAAP: FASB, rules-based | IFRS: IASB, principles-based | Korea: K-IFRS
- Same purpose; presentation order flipped (GAAP current items first, IFRS non-current first)

2.5 Taxes

- Average = total tax ÷ taxable income | Marginal = rate on the next $1 (use for decisions)
- Flat tax → marginal = average; progressive → marginal > average
- US flat 21% (2017 reform; before: progressive up to 35%) | Korea progressive 10/20/22/25%
- OECD Pillar Two: 15% minimum effective tax for large MNEs (≥ €750M)
- Ex: Korea ₩500M: 10% × 200M + 20% × 300M = ₩80M → average **16%**, marginal **20%**

2.6 Cash Flow (CFFA)

- **CFFA = OCF − NCS − ΔNWC = CF to creditors + CF to stockholders**
- OCF = EBIT + Depreciation − Taxes

- NCS = End net fixed assets − Beg net fixed assets + Depreciation
- ΔNWC = End NWC − Beg NWC
- CF/CR = Interest − Net new borrowing | CF/SH = Dividends − Net new equity
- Ex: OCF = 670 + 89 − 126 = 633 | NCS = 1,709 − 1,644 + 89 = 154 | ΔNWC = (1,403 − 389) − (1,112 − 428) = 330
- Ex: CFFA = 633 − 154 − 330 = **149** | check: CF/CR 70 − 46 = 24, CF/SH 165 − 40 = 125, sum 149
- Ex (Zephyr): 3,180 − 1,400 − 600 = **1,180**
- Traps: depreciation is added back (OCF, NCS) | current-liability increase lowers NWC | negative CF/CR = borrowing > interest paid | omit any component → wrong
- Cash flow ≠ accounting statement of cash flows

Ch.3 Working with Financial Statements

3.1 Common-Size Statements; Users

- Balance sheet ÷ total assets | income statement ÷ sales; compare across years and across different-size firms
- Ex: inventory 84,600 ÷ 470,000 = **18.0%**
- Internal users: evaluation, planning, control | External: lenders, suppliers, customers, investors, competitors / acquirers / regulators

3.2 Ratio Categories

- Liquidity (short-term solvency) | Leverage (long-term solvency) | Turnover (asset management) | Profitability | Market value
- Slide firm: Sales 2,361; NI 474; TA 3,630; TE 2,625; price $115; 33M shares

3.3 Liquidity

- Current = CA ÷ CL → **1.29**
- Quick = (CA − Inventory) ÷ CL → **0.52** (excludes inventory)
- Cash = Cash ÷ CL → **0.18**

3.4 Leverage and Coverage

- Total debt ratio = (TA − TE) ÷ TA → **0.28**
- D/E = TD ÷ TE → **0.38** | EM = TA ÷ TE = 1 + D/E → **1.38**
- TIE = EBIT ÷ Interest = 741 ÷ 141 → **5.26**
- Cash coverage = (EBIT + Dep.) ÷ Interest = 1,017 ÷ 141 → **7.21** (always ≥ TIE)

3.5 Turnover

- Inventory = **COGS** ÷ Inventory → 3.18 (numerator is COGS, not sales)
- Receivables = Sales ÷ AR → 12.56 | Payables = COGS ÷ AP → 3.91
- Total asset turnover (TAT) = Sales ÷ TA → 0.65

3.6 Profitability

- PM = NI ÷ Sales → 20.08% | ROA = NI ÷ TA → 13.06% | ROE = NI ÷ TE → 18.06%
- Same numerator; only the denominators differ

3.7 Market Value

- EPS = NI ÷ shares → 14.36 | P/E = price ÷ EPS → 8.01
- P/S = price ÷ (Sales/share) → 1.61 | M/B = price ÷ (TE/share) → 1.45 (book value per share)
- EV = market cap + debt − cash; EBITDA = EBIT + Dep.
- Ex: EV = 3,795 + 1,005 − 98 = 4,702; EBITDA = 1,017; **EV/EBITDA = 4.62**

3.8 DuPont

- **ROE = PM × TAT × EM** (a product, not a sum) | PM = profitability / cost control, TAT = asset efficiency, EM = leverage
- Ex: Coca-Cola 27.34% × 0.46 × 3.26 ≈ **40.7%** (slide value; inputs rounded)
- KO vs PEP: both ROE ≈ 40%, different paths
  - KO: sells concentrate, asset-light, high margin, low turnover
  - PEP: finished goods and snacks, ~2× sales, thin margin, high turnover and leverage
  - PEP current ratio < 1 is not a red flag (cash sales, long payables terms, strong credit)

3.9 Benchmarks and Limits

- Benchmarks: time-trend analysis; peer group (SIC / NAICS)
- Limits: no underlying theory, conglomerates, global (IFRS vs GAAP), accounting methods (FIFO / LIFO, depreciation), different year-ends, capital structure, seasonality / one-offs, window dressing

Ch.4 Time Value of Money

4.1 Concepts

- A dollar today > a dollar tomorrow (it can be invested)
- Compounding (interest on interest) ↔ discounting
- Simple = interest on principal only | compound = on principal + past interest; gap grows with time
- Ex: $1,000, 10%, 3 yrs: simple 1,000 + 3 × 100 = 1,300 vs compound 1,000 × 1.10^3 = **1,331** (interest-on-interest $31)

4.2 Formulas

- $FV = PV(1 + r)^t$ | $PV = FV \div (1 + r)^t$
- $r = (FV \div PV)^{1/t} - 1$ | $t = \ln(FV \div PV) \div \ln(1 + r)$
- Slides favor `CPT I/Y` and `CPT N` over memorizing r and t formulas

4.3 Keystrokes

- FV: $2,000, 8%, 5 yrs → `5 N` `8 I/Y` `0 PMT` `2000 +/− PV` `CPT FV` = **2,938.66**
- PV: $10,000 in 6 yrs, 7% → `6 N` `7 I/Y` `0 PMT` `10000 FV` `CPT PV` = **−6,663.42**
- r: $5,000 → $10,000 in 9 yrs → `9 N` `5000 +/− PV` `0 PMT` `10000 FV` `CPT I/Y` = **8.01%**
- t: $1,000 → $2,500 at 9% → `9 I/Y` `1000 +/− PV` `0 PMT` `2500 FV` `CPT N` = **10.63 yrs**
- Practice (a) FV of $8,000, 6.5%, 10 yrs → 15,017.10
- Practice (b) PV of $20,000 in 10 yrs at 6.5% → −10,654.52
- Practice (c) $8,000 → $20,000 in 10 yrs, r → 9.60%
- Practice (d) $8,000 → $20,000 at 6.5%, t → 14.55 yrs
- Trap: same sign on PV and FV → Error 5

Ch.5 Discounted Cash Flow Valuation

5.1 Multiple Cash Flows

- Discount / compound each cash flow at its own t, then **add** (never average); including CF0 → NPV
- FV: $4,000 now + $7,000 in 1 yr, 8%, value at year 3
  - `3 N` `8 I/Y` `0 PMT` `4000 +/− PV` `CPT FV` = 5,038.85
  - `2 N` `7000 +/− PV` `CPT FV` = 8,164.80 → sum = **13,203.65**
- PV of uneven CFs 200 / 400 / 600 / 800 at 12%
  - `CF` `2nd CLR WORK` `0 ENTER ↓` (CF0) `200 ENTER ↓↓` `400 ENTER ↓↓` `600 ENTER ↓↓` `800 ENTER ↓↓`
  - `NPV` `12 ENTER ↓` `CPT` = **1,432.93**

5.2 Annuity Types and Formulas

- Annuity: level CFs, finite (car / home loans) | Annuity due: paid at start (rent, lease, insurance) → × (1 + r)
- Perpetuity: C ÷ r (preferred stock, consol) | Growing perpetuity: $C_1 \div (r - g)$, r > g
- $PV = C \times [1 - 1/(1 + r)^t] \div r$ | $FV = C \times [(1 + r)^t - 1] \div r$
- r = rate **per payment period** (not APR); t = **number of payments** (not years)

5.3 Annuity Keystrokes

- PV (loan amount): $632 / mo, 1% / mo, 48 mo → `48 N` `1 I/Y` `632 +/− PMT` `0 FV` `CPT PV` = **23,999.54**
- Practice: $450, 0.75%, 36 mo → 14,151.06
- PMT: $20,000, 0.5% / mo, 60 mo → `60 N` `0.5 I/Y` `20000 PV` `0 FV` `CPT PMT` = **−386.66**
- Practice: $15,000, APR 9% monthly, 4 yrs → r = 0.75%, N = 48 → −373.28
- r: cost $1,000, $200 / yr × 8 → `8 N` `1000 +/− PV` `200 PMT` `0 FV` `CPT I/Y` = **11.81%**
- Practice: $8,000, $1,200 / yr × 12 → 10.45%
- N: save $2,000 / yr at 7% to reach $50,000 → `7 I/Y` `2000 +/− PMT` `0 PV` `50000 FV` `CPT N` = **14.95 yrs**
- Practice: $400 / mo, 0.5% / mo, $30,000 → 63.85 mo (5.32 yrs)
- FV: $300 / mo, 0.75% / mo, 30 yrs → `360 N` `0.75 I/Y` `300 +/− PMT` `0 PV` `CPT FV` = **549,223.04**
- Practice: $2,500 / yr, 7%, 25 yrs → 158,122.59

5.4 Annuity Due

- Method 1: ordinary value × (1 + r) | Method 2: `2nd BGN` `2nd SET` `2nd QUIT`, enter as usual, then **restore END**
- Ex: $1,000 / mo, 1% / mo, 36 mo → ordinary PV 30,107.51 × 1.01 = **30,408.58**
- Practice: $1,800, 0.5%, 24 mo → 40,613.16 × 1.005 = 40,816.22
- Lease PMT (BGN): $48,000, 12% APR monthly, 4 yrs → `48 N` `1 I/Y` `48000 +/− PV` `0 FV` `CPT PMT` = **1,251.51** (= ordinary 1,264.02 ÷ 1.01)

5.5 Perpetuities

- PV = C ÷ r | Ex: preferred $1.50 / qtr at 2% / qtr → **75.00**
- Growing: $C_1 \div (r - g)$ | Ex: $2.00 next year, g 3%, r 10% → 2.00 ÷ 0.07 = **28.57**
- If $2.00 was just paid ($D_0$): $C_1 = D_0(1 + g)$ = 2.06 → 2.06 ÷ 0.07 = **29.43**

5.6 APR vs. EAR

- APR = quoted rate (periodic rate × m) | EAR = actual annual rate | **always compare in EAR**
- EAR = $(1 + APR/m)^m - 1$ | APR = $m \times [(1 + EAR)^{1/m} - 1]$ | continuous: EAR = $e^r - 1$
- More frequent compounding / higher rate → larger APR–EAR gap
- `2nd ICONV`: NOM → C/Y → `CPT EFF`; reverse: EFF → C/Y → `CPT NOM` (standard BA II Plus operation)
- Ex: 15% APR → annual 15.00%, semiannual 15.56%, quarterly 15.87%, monthly 16.08%, daily 16.18%
- Practice: 9% APR → quarterly 9.31%, monthly 9.38%, daily 9.42%
- Bank A 15% daily (EAR 16.18%) vs Bank B 15.5% annual (15.50%) → A for deposits
- Card A 22.9% monthly (EAR 25.46%) vs Card B 23.5% annual (23.50%) → B is cheaper
- EAR 12% as monthly APR → **11.39%** | EAR 9% as quarterly APR → **8.71%**
- Calculator returns a monthly rate: $250 / mo × 15 yrs → $75,000 → `180 N` `250 +/− PMT` `0 PV` `75000 FV` `CPT I/Y` = 0.5303% / mo → APR **6.36%**, EAR **6.55%**

5.7 Loan Types and Amortization

- Pure discount (T-bill, zero) | interest-only (corporate bond) | amortized (home / car loan)
- Loan balance = PV of remaining payments
- Payment constant; interest falls, principal rises
- Interest = beginning balance × r | principal = PMT − interest | ending = beginning − principal
- Ex: $5,000, 9%, 5 yrs → `5 N` `9 I/Y` `5000 PV` `0 FV` `CPT PMT` = **−1,285.46**
  - Year 1: interest 450.00, principal 835.46, ending balance 4,164.54
  - Total paid 6,427.30 = principal 5,000 + interest 1,427.30
- Practice: $10,000, 8%, 4 yrs → PMT 3,019.21 | yr-1 ending balance 7,780.79 | yr-2 interest 622.46 | balance after 2 yrs 5,384.05

Ch.6 Interest Rates and Bond Valuation

6.1 Bond Anatomy and Keys

- Face / par = repaid at maturity | coupon = fixed interest | coupon rate = fixed at issue | maturity
- YTM = market required return, changes daily | after issue only the **price** changes
- Price = PV(coupon annuity) + PV(face)
- Keys: N = periods to maturity | I/Y = YTM per period | PMT = coupon per period | FV = face | PV = price
- PMT and FV positive → PV comes out negative (cash you pay); solving YTM → enter PV as negative

6.2 Par, Discount, Premium

- Coupon rate = YTM → **par**
- Coupon rate < YTM → **discount** (capital gain at maturity)
- Coupon rate > YTM → **premium** (capital loss at maturity)

6.3 Pricing Keystrokes

- Discount: $1,000, 8% coupon, 10 yrs, YTM 10% → `10 N` `10 I/Y` `80 PMT` `1000 FV` `CPT PV` = **−877.11**
- Premium: same bond, YTM 6% → `10 N` `6 I/Y` `80 PMT` `1000 FV` `CPT PV` = **−1,147.20**
- Semiannual: 9% coupon, 7 yrs, YTM 8% → N = 2 × yrs, I/Y = YTM ÷ 2, PMT = coupon ÷ 2
  - `14 N` `4 I/Y` `45 PMT` `1000 FV` `CPT PV` = **−1,052.82**
- Zero coupon: 5 yrs, YTM 6% semiannual → `10 N` `3 I/Y` `0 PMT` `1000 FV` `CPT PV` = **−744.09**

6.4 Yield to Maturity

- YTM = single discount rate that sets PV of promised CFs = price (the bond's IRR)
- Assumptions: hold to maturity | no default | coupons reinvested at YTM
- Annual: 8% coupon, 8 yrs, price 918.50 → `8 N` `918.50 +/− PV` `80 PMT` `1000 FV` `CPT I/Y` = **9.50%**
- Semiannual: 7% coupon, 10 yrs, price 900.29 → `20 N` `900.29 +/− PV` `35 PMT` `1000 FV` `CPT I/Y` = 4.25 → × 2 = **8.50%** (EAY 8.68%)
- Three yields (8% coupon, price 918.50): coupon rate 8.00% < current yield 80 ÷ 918.50 = 8.71% < YTM 9.50%

6.5 Interest Rate Risk

- Price and yield move **inversely** (promised CFs are fixed); price curve is convex
- Longer maturity and lower coupon → more sensitive; zero coupon is most sensitive
- Interest rate risk = price risk (not default risk)
- 30-yr bond, yield 10% → 12%: coupon 5% → −17.50% | 10% → −16.11% | 15% → −15.61%

6.6 Clean vs. Dirty Price

- Dirty (invoice) = clean (quoted) + accrued interest
- Accrued = coupon × (elapsed ÷ coupon period)
- Ex: semiannual $40 coupon, 2 months elapsed, quote $985 → 40 × 2/6 = 13.33 → dirty **998.33**

6.7 Fisher Effect and Yield Components

- **1 + R = (1 + r)(1 + h)** | Ex: nominal 15.5%, inflation 5% → 1.155 ÷ 1.05 − 1 = **10.00%** real (R − h = 10.5% is only approximate)
- Nominal yield = real rate + inflation premium + interest-rate-risk premium (rises with maturity) + bond-specific: default, taxability, liquidity premiums
- Higher expected inflation → higher nominal yields, lower bond prices

6.8 Term Structure

- Relation between maturity and yield (pure-discount Treasuries); yield curve = its graph
- Upward sloping → expected inflation up | downward → expected inflation down

6.9 Treasuries and Municipals

- Treasury: no default risk, state / local tax-exempt, federally taxable
- Municipal: default risk, federally tax-exempt → lower yield, held by high-bracket investors
- Tax-equivalent yield = muni yield ÷ (1 − t) | 6 ÷ 0.65 = **9.23%** (t = 35%) | 6 ÷ 0.80 = **7.50%** (t = 20%)

6.10 Debt vs. Equity

- Creditors: no ownership, no vote | interest is a pre-tax (deductible) expense | dividends are paid after tax
- Unpaid interest → default; unpaid dividend → not default
- Liquidation: equity is the residual claim (last)

6.11 Indenture, Call, Covenants

- Indenture = contract between issuer and bondholders; trustee represents holders
- Terms: face, coupon, maturity | security | repayment (sinking fund) | seniority | call | covenants
- Call: issuer may repurchase before maturity | call premium = call price − face | deferred call = protected period
  - Exercised when rates fall → bond value lower; investors demand a higher yield
- Negative covenants: limit dividends, pledging assets, mergers, asset sales, extra long-term debt
- Positive covenants: maintain NWC, supply audited statements, maintain collateral, keep ratios
- Firms accept covenants because the yield reduction exceeds the flexibility cost

6.12 Ratings and Bond Types

- Investment grade AAA / Aaa to BBB / Baa | junk (speculative) BB / Ba and below
- BBB− / Baa3 = last investment grade (institutional-investment cutoff)
- Zero coupon: only CF at maturity | Floating rate: collar = floor + cap | Convertible: lower coupon
- Put provision: opposite of call, adds value | CoCo | Sukuk: asset-based return instead of interest

6.13 Bond Market

- Large volume but mostly **OTC** (no central exchange); dealer quotes; thin liquidity per issue; institution-dominated
- Transparency problem → TRACE (2002) reporting and disclosure

Ch.7 Equity Markets and Stock Valuation

7.1 Stock vs. Bond Valuation

- Dividends at board discretion (coupons are contractual)
- No maturity → infinite cash flows
- Must estimate R and g; price is highly sensitive to both

7.2 Dividend Discount Model

- P_0 = PV of all future dividends
- "Sell later at a higher price" is already inside the model
- Non-dividend firms still have value only if they eventually pay
- No slide keystrokes for Ch.7 → solve with formulas

7.3 Four Growth Cases

- **Zero growth** (preferred): $P_0 = D \div R$
- **Constant growth**: $D_1 = D_0(1 + g)$; $P_0 = D_1 \div (R - g)$, requires $R > g$; $P_t = D_{t+1} \div (R - g)$
- **Nonconstant**: list $D_1 \ldots D_t$, find P_t, discount all; P_t is discounted t periods
- **Two-stage** (g_1 for t yrs, then g_2): $P_0 = D_1/(R - g_1) \times [1 - ((1 + g_1)/(1 + R))^t] + P_t/(1 + R)^t$, with $P_t = D_{t+1}/(R - g_2)$ (standard form; slide formula is an image)
- Sustainable growth is low single digits
- Zero: D \$3.00, R 10% → **30.00**
- Constant: D_0 \$2.00, g 5%, R 12% → D_1 = 2.10; 2.10 ÷ 0.07 = **30.00**
- Future price: $D_6 = 2.00 \times 1.05^6$; $D_6 \div 0.07$ = **38.29**
- Nonconstant: D_0 \$1.00, 20% growth for 3 yrs then 5% forever, R 15%
  - $D_{1\ldots3}$ = 1.20, 1.44, 1.728 | D_4 = 1.8144 | P_3 = 1.8144 ÷ 0.10 = 18.144
  - $P_0 = 1.20/1.15 + 1.44/1.15^2 + 1.728/1.15^3 + 18.144/1.15^3$ = **15.20**
- No dividend for 4 yrs, \$0.50 at end of yr 5, g 4%, R 20% → $P_4 = 0.50 \div (0.20 - 0.04) = 3.125$; $\div 1.20^4$ = **1.51**

7.4 Return Decomposition

- **R = dividend yield ($D_1 \div P_0$) + capital gains yield (= g under constant growth)**
- Dividend yield is like a bond's current yield
- Ex: P \$20, D_1 \$1, g 10% → 5% + 10% = **15%**

7.5 Multiples

- P/E, price-sales: fallback when there is no dividend
- Benchmark = comparable firms or own history; "comparable" drives the answer
- Losses → P/E meaningless; P/S still works
- Price = EPS × benchmark P/E | Ex: 3.00 × 18 = **54.00**
- Price = sales per share × benchmark P/S | Ex: 10.00 × 2.5 = **25.00**

7.6 Common Stock Rights

- Elect directors; vote (mergers, charter amendments, new shares)
- Dividends only after board declaration | residual claim | limited liability (loss up to paid-in)
- Preemptive right: first claim on new shares, prevents dilution
- Proxy: delegated vote (management usually favored); proxy fight: rival solicits proxies
- Dual-class: Class B 10 votes vs Class A 1 (founder control); one-share-one-vote varies by law / listing rules

7.7 Dividends

- Discretionary (non-payment ≠ default)
- Paid from after-tax income (asymmetric to deductible interest)
- US corporate dividends-received deduction (DRD) → corporations prefer preferred stock

7.8 Preferred Stock

- Fixed dividend on par; paid before common, after creditors
- Cumulative: arrearages paid first | normally no vote (may gain vote in arrears)
- Looks like debt but is not: missed dividend ≠ bankruptcy trigger; dividends not tax-deductible; equity legally and in accounting

7.9 Stock Markets

- Primary (firm receives funds) vs secondary (firm receives nothing)
- Dealer: holds inventory, bears risk | broker: agent, earns commission
- Bid = dealer buys | ask = dealer sells | spread = dealer's compensation
- NYSE: hybrid (DMM, mostly electronic) | NASDAQ: dealer market, competing market makers, ECNs narrow spreads
- Quote fields: Last / Close, Net change, Bid / Ask, Volume, Dividend, Yield = dividend ÷ price, P/E = price ÷ EPS

Exam Trap Checklist

T.1 Calculator

- Same sign on PV and FV → Error 5; cash out −, cash in +
- `2nd CLR TVM` / `2nd CLR WORK` before each problem; P/Y = 1
- Match units of N and I/Y (semiannual: N = yrs × 2, I/Y = rate ÷ 2)
- First CF now → annuity due = ordinary × (1 + r); turn BGN off afterwards

T.2 Statements and Ratios

- Interest deducted after EBIT | NCS adds depreciation | OCF adds depreciation back
- Inventory turnover uses COGS | quick ratio excludes inventory | EM = 1 + D/E | M/B uses book value per share
- DuPont is a product, not a sum

T.3 Rates and Bonds

- APR given → convert to the periodic rate first; EAR = $(1 + APR/m)^m - 1$; compare in EAR
- After-tax interest = interest × (1 − T); dividends paid from after-tax income
- Price and YTM move inversely; coupon rate < YTM → discount; clean + accrued = dirty
- Tax-equivalent yield: **divide** by (1 − t) (6% × 0.65 = 3.90% is wrong)

T.4 Stocks and Written Answers

- Constant growth numerator is D_1 ($D_0 \times (1 + g)$); valid only if R > g
- P_3 is discounted 3 periods, not 4; losses → use P/S
- Written answers: definition + direction (up / down) + reason in one line

BUSINESS FINANCE — FINANCIAL RATIOS

Chapter 3 | One-page formula sheet · Use figures from the same fiscal period where possible

1. COMMON-SIZE STATEMENTS

Ratio / purpose	Formula
Balance sheet account (%)	Account ÷ Total Assets × 100
Income statement item (%)	Item ÷ Sales × 100

2. LIQUIDITY — short-term solvency

Ratio / purpose	Formula
Current ratio	Current Assets (CA) ÷ Current Liabilities (CL)
Quick ratio	(CA – Inventory) ÷ CL
Cash ratio	Cash ÷ CL

3. LEVERAGE / LONG-TERM SOLVENCY

Ratio / purpose	Formula
Total debt ratio	(Total Assets – Total Equity) ÷ Total Assets = Total Debt ÷ TA
Debt-to-equity (D/E)	Total Debt ÷ Total Equity
Equity multiplier (EM)	Total Assets ÷ Total Equity = 1 + D/E
Times interest earned (TIE)	EBIT ÷ Interest
Cash coverage	(EBIT + Depreciation) ÷ Interest

4. ASSET MANAGEMENT / TURNOVER

Ratio / purpose	Formula
Inventory turnover	COGS ÷ Inventory
Receivables turnover	Sales ÷ Accounts Receivable (AR)
Payables turnover	COGS ÷ Accounts Payable (AP)
Total asset turnover (TAT)	Sales ÷ Total Assets

5. PROFITABILITY

Ratio / purpose	Formula
Profit margin (PM)	Net Income (NI) ÷ Sales
Return on assets (ROA)	NI ÷ Total Assets
Return on equity (ROE)	NI ÷ Total Equity

6. MARKET VALUE

Ratio / purpose	Formula
Earnings per share (EPS)	Net Income ÷ Shares Outstanding
P/E ratio	Price per Share ÷ EPS
Price/Sales (P/S)	Price per Share ÷ Sales per Share
Market-to-book (M/B)	Price per Share ÷ Book Value per Share
Enterprise value (EV)	Market Value of Equity + Book Value of Liabilities – Cash
EBITDA	EBIT + Depreciation & Amortization
EV/EBITDA	Enterprise Value ÷ EBITDA

7. DUPONT IDENTITY — decompose ROE

Ratio / purpose	Formula
Core identity	ROE = PM × TAT × EM
Expanded	(NI ÷ Sales) × (Sales ÷ TA) × (TA ÷ Equity) = NI ÷ Equity

QUICK INTERPRETATION & EXAM CHECKS

Liquidity: higher ratios usually mean more short-term coverage, but excess idle assets may reduce efficiency. • **Leverage:** more debt can magnify shareholder returns and risk; TIE / cash coverage measure ability to cover interest. • **Turnover:** higher turnover generally means more efficient asset use; interpret inventory turnover in industry context. • **Profitability:** PM = profit per sales dollar; ROA = profit relative to assets; ROE = profit relative to equity. • **Common-size:** balance sheet uses total assets as denominator; income statement uses sales. • **DuPont:** ROE can come from high margin, efficient asset turnover, or high leverage. • **Calculation:** convert percentages to % only at the end; ratios/turnover are often shown as “x” (e.g., 1.5x). • **Comparison cautions:** compare across time and similar firms; accounting methods, seasonality, fiscal year-end, one-offs, and window dressing can mislead.

Chapter 2: Financial Statements, Taxes and Cash Flow

Multiple Choice

1. A firm has net sales of $2,400, cost of goods sold of $1,300, depreciation of $200 and interest paid of $150. The tax rate is 21%. What is net income?
a. $900.00
b. $711.00
c. $592.50
d. $750.00

2. A Korean firm has taxable income of ₩500 million. Using the national corporate tax schedule in the table, what are its average and marginal tax rates?

Taxable income	Rate
Up to ₩200 million	10%
₩200 million to ₩20 billion	20%
₩20 billion to ₩300 billion	22%
Over ₩300 billion	25%

a. Average 15%; marginal 20%
b. Average 16%; marginal 20%
c. Average 16%; marginal 10%
d. Average 20%; marginal 20%

3. A firm issues 2,000 shares of common stock with a par value of $0.50 at an issue price of $12 per share. How much goes to additional paid-in capital (paid-in surplus)?
a. $1,000
b. $24,000
c. $22,000
d. $23,000

4. Zephyr Logistics, 2025: EBIT $2,600, depreciation $1,000, taxes $420. Net fixed assets rose from $9,000 to $9,400. Current assets rose from $5,000 to $5,800 and current liabilities rose from $2,500 to $2,700. What is the cash flow from assets (CFFA)?
a. $1,180
b. $1,780
c. $2,380
d. $3,180

5. A firm paid interest of $100 and dividends of $885. Its long-term debt rose from $3,000 to $3,200 and its common stock and paid-in surplus rose from $2,500 to $2,600 because of a new share issue. What are its cash flow to creditors and cash flow to stockholders?

a. −$100 and $785

b. $300 and $985

c. $100 and $885

d. $785 and −$100

Chapter 2: Financial Statements, Taxes and Cash Flow — Answers

1. c
EBIT = 2,400 − 1,300 − 200 = 900; taxable income = 900 − 150 = 750; taxes = 21% × 750 = 157.50; net income = 750 − 157.50 = $592.50.

2. b
Tax = 10% × 200M + 20% × 300M = ₩80M, so the average rate = 80 / 500 = 16%; the marginal rate is the 20% bracket that applies to the next won of income.

3. d
Paid-in surplus = (issue price − par) × shares = (12 − 0.50) × 2,000 = $23,000; the $1,000 of par goes to the common stock line.

4. a
OCF = 2,600 + 1,000 − 420 = 3,180; NCS = 9,400 − 9,000 + 1,000 = 1,400; ΔNWC = (5,800 − 2,700) − (5,000 − 2,500) = 600; CFFA = OCF − NCS − ΔNWC = 3,180 − 1,400 − 600 = $1,180.

5. a
CF to creditors = interest paid − net new borrowing = 100 − 200 = −$100; CF to stockholders = dividends paid − net new equity raised = 885 − 100 = $785; their sum, $685, equals CFFA.

Chapter 3: Working with Financial Statements

Multiple Choice

1. Coastal Furniture has inventory of $84,600, equity of $351,000, total assets of $470,000 and sales of $612,300. What is the common-size percentage for inventory?
a. 74.7%
b. 24.1%
c. 18.0%
d. 13.8%

2. A firm has cash of $98, current assets of $708 (including inventory of $422) and current liabilities of $548. What is its quick ratio?
a. 0.18
b. 0.52
c. 1.29
d. 0.77

3. A firm has total assets of $3,630 and total equity of $2,625. What are its debt/equity ratio and its equity multiplier?
a. D/E = 1.38; EM = 0.38
b. D/E = 0.28; EM = 1.28
c. D/E = 0.28; EM = 0.38
d. D/E = 0.38; EM = 1.38

4. A firm has EBIT of $741, depreciation of $276 and interest of $141. What are its times interest earned (TIE) and cash coverage ratios?
a. TIE = 3.30; cash coverage = 5.26
b. TIE = 7.21; cash coverage = 5.26
c. TIE = 5.26; cash coverage = 3.30
d. TIE = 5.26; cash coverage = 7.21

5. A firm has sales of $2,361, cost of goods sold of $1,344, inventory of $422 and accounts receivable of $188. What is its inventory turnover?
a. 3.18
b. 0.31
c. 7.15
d. 5.59

6. A firm has net income of $474, sales of $2,361, total assets of $3,630 and total equity of $2,625. What are its profit margin, ROA and ROE?
a. 18.06%, 13.06%, 20.08%
b. 13.06%, 20.08%, 18.06%
c. 20.08%, 13.06%, 18.06%
d. 20.08%, 18.06%, 13.06%

7. A firm has net income of $474 million, total equity of $2,625 million and 33 million shares outstanding. The market price is $115 per share. What are its P/E ratio and market-to-book ratio?
a. P/E = 14.36; M/B = 79.55
b. P/E = 8.01; M/B = 1.45
c. P/E = 1.45; M/B = 8.01
d. P/E = 8.01; M/B = 0.69

8. A firm has 33 million shares at $115 each, total liabilities (book value) of $1,005 million, cash of $98 million, EBIT of $741 million and depreciation of $276 million. What is its enterprise value / EBITDA ratio?
a. 4.72
b. 4.62
c. 3.73
d. 6.35

9. Northwind: sales $5,000M, net income $1,050M, total assets $5,000M and total equity $2,500M. Using the DuPont identity, what is its ROE?
a. 21.0%
b. 10.5%
c. 31.5%
d. 42.0%

Chapter 3: Working with Financial Statements — Answers

1. c
Common-size balance sheet accounts are divided by total assets: 84,600 / 470,000 = 18.0%.

2. b
Quick ratio = (current assets − inventory) / current liabilities = (708 − 422) / 548 = 0.52.

3. d
Total debt = 3,630 − 2,625 = 1,005; D/E = 1,005 / 2,625 = 0.38; EM = TA / TE = 3,630 / 2,625 = 1.38 = 1 + D/E.

4. d
TIE = EBIT / interest = 741 / 141 = 5.26; cash coverage = (EBIT + depreciation) / interest = 1017 / 141 = 7.21.

5. a
Inventory turnover = COGS / inventory = 1,344 / 422 = 3.18 times; a higher turnover generally indicates better performance.

6. c
Profit margin = NI / sales = 474 / 2,361 = 20.08%; ROA = NI / TA = 474 / 3,630 = 13.06%; ROE = NI / TE = 474 / 2,625 = 18.06%.

7. b
EPS = 474 / 33 = 14.36, so P/E = 115 / 14.36 = 8.01; book value per share = 2,625 / 33 = 79.55, so M/B = 115 / 79.55 = 1.45.

8. b
EV = 3,795 + 1,005 − 98 = 4,702; EBITDA = 741 + 276 = 1,017; EV/EBITDA = 4,702 / 1,017 = 4.62.

9. d
PM = 1,050 / 5,000 = 21%; TAT = 5,000 / 5,000 = 1.00; EM = 5,000 / 2,500 = 2.00; ROE = PM × TAT × EM = 0.21 × 1.00 × 2.00 = 42.0%.

Chapter 4: Introduction to Valuation — Time Value of Money

Multiple Choice

1. You invest $1,000 for 3 years at 10% per year. How much more do you earn with compound interest than with simple interest?
a. $31
b. $331
c. $100
d. $0

2. You invest $2,000 today at 8% per year, compounded annually. How much will you have after 5 years?
a. $1,361.17
b. $2,800.00
c. $2,938.66
d. $2,720.98

3. You want $10,000 in 6 years and can earn 7% per year, compounded annually. How much must you invest today?
a. $7,042.25
b. $15,007.30
c. $6,663.42
d. $9,345.79

4. An investment of $5,000 grows to $10,000 in 9 years. What annual rate did it earn?
a. 7.18%
b. 8.01%
c. 9.05%
d. 11.11%

5. How long does it take for $1,000 to grow to $2,500 at 9% per year, compounded annually?
a. 10.18 years
b. 16.67 years
c. 11.63 years
d. 10.63 years

6. An investment of $18,000 today will be worth $31,000 in 7 years. Assuming P/Y = 1 and cleared TVM registers, which BA II Plus sequence returns the annual rate of return?

a. 7 N → 18,000 +/− PV → 31,000 FV → 0 PMT → CPT I/Y

b. 7 N → 31,000 +/− PV → 18,000 FV → 0 PMT → CPT I/Y

c. 7 N → 18,000 PV → 31,000 FV → 0 PMT → CPT I/Y

d. 7 N → 18,000 +/− PV → 31,000 FV → 0 PMT → CPT N

Chapter 4: Introduction to Valuation — Time Value of Money — Answers

1. a
Simple: 1,000 + 3 × 100 = 1,300; compound: 1,000 × $(1.10)^3$ = 1,331; the extra $31 is interest on interest.

2. c
FV = 2,000 × $(1.08)^5$ = $2,938.66. BA II Plus: 2nd [CLR TVM], 5 N, 8 I/Y, 0 PMT, 2000 +/− PV, CPT FV.

3. c
PV = 10,000 / $(1.07)^6$ = $6,663.42. BA II Plus: 6 N, 7 I/Y, 0 PMT, 10000 FV, CPT PV gives −6,663.42 (the cash you pay in).

4. b
r = $(10{,}000 / 5{,}000)^{1/9} − 1$ = 8.01%. BA II Plus: 9 N, −5000 PV, 0 PMT, 10000 FV, CPT I/Y.

5. d
2,500 = 1,000 × $(1.09)^t$, so t = ln(2.5) / ln(1.09) = 10.63 years. BA II Plus: 9 I/Y, −1000 PV, 0 PMT, 2500 FV, CPT N.

6. a
Cash paid out is negative and cash received is positive, so PV = −18,000 and FV = +31,000; with N = 7 and PMT = 0, CPT I/Y returns 8.08 (% per year).

Chapter 5: Discounted Cash Flow Valuation

Multiple Choice

1. You deposit $4,000 today and another $7,000 at the end of year 1. The account earns 8% per year. How much is in the account at the end of year 3?
a. $13,203.65
b. $11,000.00
c. $13,856.83
d. $12,225.60

2. An investment pays $200, $400, $600 and $800 at the end of years 1 through 4. The required return is 12%. What is it worth today?
a. $1,604.88
b. $1,271.04
c. $1,432.93
d. $2,000.00

3. You can afford a car payment of $632 per month for 48 months, and the bank quotes 1% per month. How large a loan can you take?
a. $24,239.54
b. $23,999.54
c. $2,466.04
d. $30,336.00

4. You finance $15,000 of a car over 4 years at 9% APR compounded monthly. What is the monthly payment?
a. $1,371.92
b. $385.84
c. $312.50
d. $373.28

5. An investment costs $1,000 today and pays $200 at the end of each year for 8 years. What annual return does it offer?
a. 20.00%
b. 7.50%
c. 6.05%
d. 11.81%

6. You deposit $400 at the end of every month into an account earning 0.5% per month. How many years until the balance reaches $30,000?
a. 5.32 years
b. 63.85 years
c. 75.00 years
d. 6.25 years

7. You put $2,500 into an index fund at the end of every year for 25 years, earning 7% per year. What is the balance at the end?
a. $169,191.18
b. $62,500.00
c. $158,122.59
d. $13,568.58

8. A two-year apartment lease requires $1,800 at the beginning of every month. The relevant rate is 0.5% per month, compounded monthly. What is the lease worth today?
a. $39,016.22
b. $40,816.22
c. $43,200.00
d. $40,613.16

9. A company has just paid a dividend of $2.00 per share. Dividends are expected to grow 3% per year forever, and investors require 10% per year. What is the share worth?
a. $20.00
b. $29.43
c. $28.57
d. $68.67

10. Bank A quotes 15% APR compounded daily; Bank B quotes 15.5% APR compounded annually. What are their EARs, and where would you rather deposit?
a. A = 16.18%; B = 16.76%; Bank B
b. A = 15.56%; B = 15.50%; Bank A
c. A = 15.00%; B = 15.50%; Bank B
d. A = 16.18%; B = 15.50%; Bank A

11. You borrow $10,000 for 4 years at 8%, repaid in equal annual payments of $3,019.21. What is the loan balance after two years?
a. $5,384.05
b. $7,780.79
c. $5,561.58
d. $3,961.58

Chapter 5: Discounted Cash Flow Valuation — Answers

1. a
FV = 4,000 × $(1.08)^3$ + 7,000 × $(1.08)^2$ = 5,038.85 + 8,164.80 = $13,203.65; the $4,000 is invested 3 years and the $7,000 only 2.

2. c
PV = 200/1.12 + 400/$(1.12)^2$ + 600/$(1.12)^3$ + 800/$(1.12)^4$ = $1,432.93; each cash flow is discounted by its own t and then added.

3. b
PV = 632 × [1 − 1/$(1.01)^{48}$] / 0.01 = $23,999.54. BA II Plus: 48 N, 1 I/Y, −632 PMT, 0 FV, CPT PV; N counts payments, not years.

4. d
r = 9% / 12 = 0.75% per month and t = 48: 15,000 = C × [1 − 1/$(1.0075)^{48}$] / 0.0075, so C = $373.28. BA II Plus: 48 N, 0.75 I/Y, 15000 PV, 0 FV, CPT PMT.

5. d
1,000 = 200 × [1 − 1/$(1 + r)^8$] / r cannot be solved by algebra; the calculator iterates: 8 N, −1000 PV, 200 PMT, 0 FV, CPT I/Y = 11.81%.

6. a
30,000 = 400 × [$(1.005)^t$ − 1] / 0.005 gives t = 63.85 months (CPT N); 63.85 / 12 = 5.32 years.

7. c
FV = 2,500 × [$(1.07)^{25}$ − 1] / 0.07 = $158,122.59. BA II Plus: 25 N, 7 I/Y, −2500 PMT, 0 PV, CPT FV.

8. b
PV = ordinary annuity × (1 + r) = 1,800 × [1 − 1/$(1.005)^{24}$] / 0.005 × 1.005 = 40,613.16 × 1.005 = $40,816.22.

9. b
C_1 is the cash flow one period from today: D_1 = 2.00 × 1.03 = 2.06; PV = D_1 / (r − g) = 2.06 / 0.07 = $29.43.

10. d
EAR(A) = $(1 + 0.15/365)^{365}$ − 1 = 16.18%; EAR(B) = 15.50% because compounding is annual; Bank A pays more. BA II Plus ICONV: NOM = 15, C/Y = 365, CPT EFF.

11. a
Balance = PV of the remaining 2 payments = 3,019.21 × [1 − 1/$(1.08)^2$] / 0.08 = $5,384.05.

Chapter 6: Interest Rates and Bond Valuation

Multiple Choice

1. A bond has a $1,000 face value, an 8% annual coupon and 10 years to maturity. Comparable bonds yield 10%. What is its price?
a. $1,000.00
b. $385.54
c. $877.11
d. $1,122.89

2. A $1,000 bond pays a 9% coupon semiannually and matures in 7 years. Comparable bonds yield 8% (quoted annual rate). What is its price?
a. $1,528.16
b. $1,000.00
c. $1,052.82
d. $1,052.06

3. A 5-year zero coupon bond with a $1,000 face value is priced to yield 6% compounded semiannually. What is its price?
a. $558.39
b. $744.09
c. $862.61
d. $747.26

4. A $1,000 bond with an 8% annual coupon and 8 years to maturity trades at $918.50. What is its yield to maturity?
a. 8.7%
b. 10.0%
c. 8.0%
d. 9.5%

5. A $1,000 bond with an 8% annual coupon and 8 years to maturity is priced at $918.50. What is its current yield?
a. 8.71%
b. 9.50%
c. 8.87%
d. 8.00%

6. A $1,000 bond pays an 8% coupon semiannually ($40 per coupon). Two months have passed since the last coupon, and the bond is quoted at $985. What is the invoice (dirty) price?

a. $998.33

b. $1,025.00

c. $985.00

d. $991.67

7. You invest $1,000 for one year at a nominal rate of 15.5%. Inflation is 5%. What is the real rate?

a. 16.28%

b. 10.50%

c. 10.00%

d. 14.76%

8. An investor in the 35% tax bracket compares a 6% municipal bond with a taxable bond. What is the equivalent taxable yield of the muni?

a. 7.50%

b. 9.23%

c. 6.00%

d. 3.90%

Chapter 6: Interest Rates and Bond Valuation — Answers

1. c
Price = $80 \times [1 - 1/(1.10)^{10}] / 0.10 + 1{,}000 / (1.10)^{10} = \$877.11$. BA II Plus: 10 N, 10 I/Y, 80 PMT, 1000 FV, CPT PV = −877.11.

2. c
Work in periods: N = 14, I/Y = 8% / 2 = 4%, PMT = 90 / 2 = 45, FV = 1,000; price = $1,052.82.

3. b
A zero pays only at maturity: price = $1{,}000 / (1.03)^{10} = \$744.09$ (N = 10, I/Y = 3).

4. d
$918.50 = 80 \times [1 - 1/(1 + r)^{8}] / r + 1{,}000 / (1 + r)^{8}$ gives r = 9.5% (9% prices at $944.65 and 10% at $893.30). BA II Plus: 8 N, −918.50 PV, 80 PMT, 1000 FV, CPT I/Y.

5. a
Current yield = annual coupon / market price = 80 / 918.50 = 8.71%; it counts the coupon but ignores the capital gain at maturity.

6. a
Accrued interest = 40 × (2 months / 6 months) = 13.33; dirty price = clean price + accrued interest = 985 + 13.33 = $998.33.

7. c
Fisher: $1 + R = (1 + r)(1 + h)$, so $r = 1.155 / 1.05 - 1 = 10.00\%$.

8. b
Equivalent taxable yield = muni yield / (1 − t) = 6% / 0.65 = 9.23%.

Chapter 7: Equity Markets and Stock Valuation

Multiple Choice

1. A company just paid a dividend of $2.00 per share. Dividends are expected to grow at 5% a year indefinitely, and investors require a 12% return. What is the stock worth?
a. $17.50
b. $28.57
c. $42.00
d. $30.00

2. A company just paid a dividend of $2.00 per share. Dividends are expected to grow at 5% a year indefinitely, and investors require a 12% return. What is the expected price five years from today, just after the fifth dividend has been paid?
a. $36.47
b. $30.00
c. $40.20
d. $38.29

3. A firm just paid a $1.00 dividend. Dividends will grow at 20% for the next three years and at 5% forever after that. The required return is 15%. What is the stock worth today?
a. $15.20
b. $14.63
c. $13.64
d. $3.27

4. A firm will pay no dividend for the next four years. It will pay $0.50 at the end of year 5, and that dividend will then grow at 4% forever. The required return is 20%. What is the stock worth today?
a. $3.12
b. $1.21
c. $1.51
d. $1.26

5. A stock sells for $20.00. Its next dividend will be $1.00, and dividends are expected to grow at 10% a year. What are its dividend yield, capital gains yield and required return?
a. 5.0%; 10.0%; 5.0%
b. 5.0%; 10.0%; 15.0%
c. 5.0%; 5.0%; 10.0%
d. 10.0%; 5.0%; 15.0%

6. A firm is expected to earn $3.00 per share next year. Comparable firms trade at 18 times earnings. What is the estimated share price?

a. $18.00

b. $54.00

c. $0.17

d. $21.00

7. A loss-making firm has sales of $10.00 per share. Comparable firms trade at 2.5 times sales. What is the estimated share price?

a. $4.00

b. $12.50

c. $0.00

d. $25.00

Chapter 7: Equity Markets and Stock Valuation — Answers

1. d
$D_1 = 2.00 \times 1.05 = 2.10$; $P_0 = D_1 / (R - g) = 2.10 / (0.12 - 0.05) = \$30.00$.

2. d
The model works from any date if you use the following period's dividend: $P_5 = D_6 / (R - g) = 2.00 \times (1.05)^6 / 0.07 = \$38.29$.

3. a
$D_1 = 1.20$, $D_2 = 1.44$, $D_3 = 1.728$, $D_4 = 1.728 \times 1.05 = 1.8144$; $P_3 = D_4 / (0.15 - 0.05) = 18.144$; PV $= 1.20/1.15 + 1.44/(1.15)^2 + 1.728/(1.15)^3 + 18.144/(1.15)^3 = \$15.20$.

4. c
Apply the model one period before the first dividend: $P_4 = 0.50 / (0.20 - 0.04) = 3.125$; $P_0 = 3.125 / (1.20)^4 = \$1.51$.

5. b
Dividend yield $= D_1 / P_0 = 1.00 / 20.00 = 5.0\%$; under constant growth the price grows at g, so the capital gains yield $= g = 10.0\%$; $R = 5.0\% + 10.0\% = 15.0\%$.

6. b
Price = EPS × benchmark P/E $= 3.00 \times 18 = \$54.00$; the answer is only as good as the word “comparable”.

7. d
Price = sales per share × benchmark price-sales ratio $= 10.00 \times 2.5 = \$25.00$; sales are always positive, so P/S still works when earnings are negative and a P/E ratio is meaningless.

Business Finance Midterm Short-Answer Quiz: Concepts and Keywords

1. **[Ch.2]** What does a balance sheet report, how are its two sides ordered, and how is net working capital defined?

2. **[Ch.2]** Why are the book value and the market value of a firm's assets often very different, and which one matters more for decisions?

3. **[Ch.2]** Compare US GAAP and IFRS in terms of who sets them, their style, and how the order of the balance sheet differs.

4. **[Ch.2]** Define the average tax rate and the marginal tax rate. Which one is used for financial and investment decisions, and how do the two rates compare under a flat tax and under a progressive tax?

5. **[Ch.2]** Define cash flow from assets (CFFA). What are its three components, and what other two cash flows must it equal in total?

6. **[Ch.3]** What are common-size balance sheets and income statements, and why are they useful?

7. **[Ch.3]** List the five categories of financial ratios and give the question each category answers.

8. **[Ch.3]** State the DuPont identity and explain what each of its three components measures.

9. **[Ch.3]** Coca-Cola and PepsiCo both earn an ROE of about 40%. Using the DuPont figures in the table, explain how each firm reaches that ROE.

Item	Coca-Cola (KO)	PepsiCo (PEP)
Profit margin	27.34%	8.77%
Total asset turnover	0.46	0.87
Equity multiplier	3.26	5.26
ROE	40.7%	40.4%

10. **[Ch.3]** How can a ratio be benchmarked, and what problems can distort a comparison of ratios across firms?

11. **[Ch.4]** Why is a dollar today worth more than a dollar tomorrow? Distinguish simple from compound interest, and compounding from discounting.

12. **[Ch.4]** State the sign convention and the clearing rule for the TI BA II Plus time-value keys, and what happens if the signs are wrong.

13. **[Ch.5]** Define an annuity, an annuity due, a perpetuity and a growing perpetuity. What condition does the growing perpetuity need, and which cash flow goes in the numerator?

14. **[Ch.5]** Distinguish the APR from the EAR. Why must quoted rates be converted before comparing them, and what happens to the EAR as compounding becomes more frequent?

15. **[Ch.5]** In a loan repaid with equal payments, how is each payment divided between interest and principal over time, and how do you find the loan balance at any date?

16. **[Ch.5]** List common mistakes when solving annuity problems on the calculator or with the formula.

17. **[Ch.6]** Name the defining terms of a bond and say which are fixed at issue and which changes. How is a bond valued?

18. **[Ch.6]** How does the relation between coupon rate and yield to maturity determine whether a bond sells at par, a premium or a discount? How do maturity and coupon size affect interest rate risk?

19. **[Ch.6]** Define yield to maturity. What three assumptions lie behind it, and how do the coupon rate, current yield and YTM rank for a discount bond?

20. **[Ch.6]** What is a bond indenture, who enforces it, and what are the basic terms it contains? Why does a call provision lower a bond's value, and why do firms accept protective covenants?

21. **[Ch.6]** What do bond ratings measure, which is the lowest investment-grade rating, and what are lower-rated bonds called?

22. **[Ch.6]** State the Fisher effect and list the components that make up a corporate bond's nominal yield.

23. **[Ch.7]** Why is a stock harder to value than a bond? What is the value of a share, and how can a firm that pays no dividend still have value?

24. **[Ch.7]** Name the four dividend growth cases and state what to watch for in each.

25. **[Ch.7]** Split a stock's required return into its two components and describe each. Under constant growth, what is the capital gains yield?

26. **[Ch.7]** When are the P/E and price-sales ratios used to value a share, and why does the word "comparable" matter?

27. **[Ch.7]** Is preferred stock debt or equity? Describe its main features.

28. **[Ch.7]** Distinguish primary from secondary markets and a dealer from a broker. What is the bid-ask spread, and how does NASDAQ differ from the NYSE?

Answers

1. [Ch.2] The balance sheet is a **snapshot** of the firm's assets and liabilities at a given **point in time**, and it satisfies assets = liabilities + stockholders' **equity**. Assets are listed in order of **decreasing liquidity**; liabilities are listed in ascending order of **when due** to be paid. **Net working capital** = current assets − current liabilities, which is usually positive for a healthy firm.
Keywords: snapshot · decreasing liquidity · when due · net working capital · equity

2. [Ch.2] Book value is the accounting value shown on the balance sheet, based on **historical cost**. **Market value** is the price at which assets, liabilities or equity can actually be bought or sold, so it reflects the **future** cash-generating ability of the assets. The two differ because historical cost does not track current values or growth prospects, and the **market value** is what matters for financial **decision** making.
Keywords: historical cost · market value · future · decision

3. [Ch.2] US GAAP is set by the **FASB** (under SEC oversight) and is more **rules-based**. IFRS (including K-IFRS in Korea) is set by the **IASB** and is more **principles-based**. Both report the same core statements; the labels and layout differ, e.g. GAAP lists current (liquid) items first while IFRS lists **non-current** items first, so the order flips.
Keywords: FASB · IASB · rules-based · principles-based · non-current

4. [Ch.2] The average tax rate = **total taxes** ÷ taxable income, the overall tax bite. The marginal tax rate is the rate paid on the **next dollar** of income, and it is the one used for **decision** making. Under a **flat** tax the marginal rate equals the average rate (U.S. 21%); under a **progressive** schedule (e.g. Korea) the marginal rate is higher than the average rate.
Keywords: total taxes · next dollar · decision · flat · progressive

5. [Ch.2] CFFA = **operating cash flow** (OCF) − **net capital spending** (NCS) − the change in **net working capital**. It shows how cash is generated by using assets and how it is paid to those who finance them, so it equals cash flow to **creditors** plus cash flow to **stockholders**. This differs from the accounting statement of cash flows.
Keywords: operating cash flow · net capital spending · net working capital · creditors · stockholders

6. [Ch.3] On a common-size balance sheet every account is a **percent of total assets**; on a common-size income statement every line item is a **percent of sales**. Standardized statements make it easy to compare **year-to-year** results and to compare companies of **different sizes**, particularly within the same industry.
Keywords: percent of total assets · percent of sales · year-to-year · different sizes

7. [Ch.3] Liquidity (short-term solvency) ratios ask whether the firm can pay its short-term bills; financial **leverage** (long-term solvency) ratios ask how much debt is used; asset management or **turnover** ratios ask how efficiently assets are used; **profitability** ratios ask how much profit is earned; and **market value** ratios (P/E, M/B, EBITDA ratio) compare the stock price with accounting numbers.
Keywords: liquidity · leverage · turnover · profitability · market value

8. [Ch.3] ROE = **profit margin** × total **asset turnover** × **equity multiplier**. Profit margin (NI/Sales) measures profitability and **cost control**; total asset turnover (Sales/TA) measures asset-use **efficiency**; the equity multiplier (TA/TE = 1 + D/E) measures financial **leverage**.
Keywords: profit margin · asset turnover · equity multiplier · leverage · efficiency · cost control

9. [Ch.3] Coca-Cola gets there through a very high **profit margin** (27.3%, an asset-light concentrate model) but low asset turnover (0.46). **PepsiCo** has a thin margin (8.8%) and relies on faster **asset turnover** (0.87) and greater financial **leverage** (equity multiplier 5.26 vs 3.26). Same destination, different routes.
Keywords: Coca-Cola · PepsiCo · profit margin · asset turnover · leverage

10. [Ch.3] A ratio means little alone. Benchmark it with **time-trend** analysis (the firm's own history) or **peer group** analysis (similar firms or industry averages). Comparisons can mislead because of different **accounting methods** (FIFO/LIFO, depreciation), different **fiscal year** ends or seasonality, **conglomerates** with no clean peer group, differences in capital structure, and **window dressing** around reporting dates; there is no underlying theory giving a single correct value.
Keywords: time-trend · peer group · accounting methods · fiscal year · conglomerate · window dressing

11. [Ch.4] A dollar today can be **invested** to **earn a return**. **Simple interest** is paid only on the original principal, while compound interest pays **interest on interest**, so value grows faster and faster. Compounding multiplies by $(1 + r)^t$ to get a **future value**; discounting divides by $(1 + r)^t$ to get a **present value**.
Keywords: earn a return · simple interest · interest on interest · present value · future value

12. [Ch.4] Clear the TVM registers before every problem (2nd CLR TVM), because old values stay in memory. Cash paid out is **negative** and cash received is positive, so PV and FV need **opposite** signs; otherwise **Error 5** appears when solving for I/Y or N. Enter four of N, I/Y, PV, **PMT**, FV and press **CPT** on the fifth; PMT = 0 for a single lump sum.
Keywords: clear · opposite · CPT · PMT · Error 5

13. [Ch.5] An annuity is a **level** cash flow for a fixed number of periods; an annuity due pays at the **beginning** of each period, so its value = ordinary annuity × (1 + r). A perpetuity is a cash flow that **never ends** (PV = C / r). A growing perpetuity (PV = C_1 / (r − g)) requires r **greater than g**, and C_1 is the cash flow **one period from today**, not the one just paid.
Keywords: level · beginning · never ends · greater than g · one period from today

14. [Ch.5] The APR is a **quoted** rate equal to the periodic rate × m; the **effective annual rate** (EAR = $(1 + APR/m)^m$ − 1) is the rate actually earned or paid over a year. Two loans with the same APR can cost different amounts, so convert both to EARs before you **compare** them. With **more frequent** compounding the EAR is **higher** for the same APR (15% APR: 15.00% annual, 16.08% monthly, 16.18% daily).
Keywords: quoted · effective annual rate · compare · more frequent · higher

15. [Ch.5] Each payment covers **interest and principal**. The payment is **constant**, but **interest falls** each period because the balance falls, so the **principal rises**. The loan balance at any date equals the **present value of the remaining payments**, discounted at the loan rate.
Keywords: interest and principal · interest falls · principal rises · constant · present value of the remaining payments

16. [Ch.5] Leaving the calculator in **BGN** mode corrupts later answers by a factor of (1 + r). The rate and N must be per payment period: use the **rate per period** (not the APR) and the **number of payments**, not the number of years. Always **clear** the TVM registers, and give PV and PMT the correct **sign**.
Keywords: BGN · rate per period · number of payments · clear · sign

17. [Ch.6] A bond's **face value** (repaid at **maturity**) and its **coupon** (coupon rate × face value) are fixed at issue. The **yield to maturity**, the market's required return, **changes** every day with the **market**. Bond value = the **present value** of the coupon annuity plus the present value of the face value, so everything that happens after issue happens to the price.
Keywords: face value · coupon · maturity · yield to maturity · present value · changes

18. [Ch.6] If the coupon rate equals the yield the bond sells at **par**; if the coupon rate exceeds the yield it sells at a **premium**; if it is below the yield it sells at a **discount**. Prices and yields move in the **opposite** direction. Interest rate risk is greater for a **longer maturity** and for a **lower coupon** (a zero coupon bond is the extreme).
Keywords: premium · discount · par · inverse · longer maturity · lower coupon

19. [Ch.6] YTM is the single discount rate that sets the present value of the promised cash flows equal to the **market price**: the bond's **internal rate of return**. It assumes the bond is **held to maturity**, there is **no default**, and coupons are **reinvested** at the YTM. For a discount bond, coupon rate < **current yield** < YTM.
Keywords: internal rate of return · market price · held to maturity · no default · reinvested · coupon rate < current yield

20. [Ch.6] The **indenture** is the written agreement between the issuer and its creditors (face value, coupon, security, seniority, repayment, call provisions, covenants); a **trustee**, usually a bank, enforces it. A **sinking fund** retires the issue gradually. A **call** provision gives the issuer the option to buy the bonds back, usually when rates have fallen, so investors demand a **higher yield** (a lower value). Protective **covenants** limit the borrower (negative and positive) and firms accept them because the reduction in yield is worth more than the lost flexibility.
Keywords: indenture · trustee · sinking fund · call · higher yield · covenant

21. [Ch.6] Ratings assess the likelihood that the promised payments are made (**default** risk) and the protection creditors have if they are not. **Investment grade** runs from AAA/Aaa down to **BBB** (Baa3 / BBB−, the lowest investment-grade rating); below that the bonds are **speculative** grade, or junk.
Keywords: default · investment grade · speculative · BBB

22. [Ch.6] The **Fisher** effect is 1 + R = (1 + r)(1 + h): nominal yields rise with expected inflation. The nominal yield = the **real rate** + **inflation premium** + **interest rate risk premium** (the term structure) plus bond-specific premiums: **default risk premium**, taxability premium and **liquidity premium**.
Keywords: Fisher · real rate · inflation premium · interest rate risk premium · default risk premium · liquidity premium

23. [Ch.7] A dividend is paid at the board's **discretion** (a bond coupon is contractual), a stock has **no maturity** date, and both the required return and the growth rate must be **estimated**. The price of a share is the **present value of all future dividends**; "I buy it to sell higher" is contained in the model. A firm that pays nothing now must **eventually** pay, or its shares are worthless.
Keywords: discretion · no maturity · estimate · present value of all future dividends · eventually

24. [Ch.7] **Zero growth** is a perpetuity (preferred stock; $D_1 = D_0$). **Constant growth** is the dividend growth model $P_0 = D_1 / (R - g)$: use the **next dividend** D_1 and require **R > g**. **Nonconstant** growth is irregular and then constant: find $P_t = D_{t+1} / (R - g)$ and discount it by t. **Two-stage** growth is g_1 for t years and then g_2.
Keywords: zero growth · constant growth · nonconstant · two-stage · next dividend · R > g

25. [Ch.7] R = **dividend yield** + **capital gains yield**. The dividend yield is **next year's dividend** divided by today's price (D_1 / P_0), the analogue of a bond's current yield. Under constant growth the price grows at exactly **g**, so the capital gains yield equals the **growth rate** g.
Keywords: dividend yield · capital gains yield · growth rate · next year's dividend

26. [Ch.7] When there is no dividend to discount, price = EPS × a **benchmark price-earnings** ratio, or sales per share × a **price-sales** ratio, taken from **comparable** firms (same industry, size and prospects) or the firm's own history. With **negative earnings** a P/E is meaningless but sales are always positive. The answer is only as good as the word comparable.
Keywords: price-earnings · price-sales · negative earnings · comparable · benchmark

27. [Ch.7] Preferred stock looks like debt (a **fixed dividend**, no voting rights, ranked **ahead of common** stock and often callable), but it is equity: skipping a dividend is **not default**, dividends are **not tax-deductible**, and holders rank **behind creditors**. Most issues are **cumulative**, so arrears must be paid before common dividends resume.
Keywords: fixed dividend · not default · cumulative · ahead of common · behind creditors · not tax-deductible

28. [Ch.7] In the **primary** market the issuer sells securities for the first time and raises money; in the **secondary** market investors trade existing shares and the firm receives nothing. A **dealer** trades for its own account and bears inventory risk; a **broker** only matches trades for a commission. The dealer's **bid** is the price it pays, the ask is the price it charges, and the **spread** is its compensation. NASDAQ is a dealer market (a computer network of competing market makers); the NYSE is a hybrid auction market with DMMs, now largely electronic.
Keywords: primary · secondary · dealer · broker · bid · spread